

5.1

Take derivative and set to zero:

$$-2X^T(y - X\beta) + 2\tau\beta = X^T X\beta - X^T y + \tau\beta = 0$$

$$\Rightarrow (X^T X + \tau I_D) \beta = X^T y$$

$$\text{We get: } \hat{\beta}_\tau = (X^T X + \tau I_D)^{-1} X^T y$$

$$\stackrel{S_\tau}{=} S_\tau^{-1} X^T y$$

$$\stackrel{y}{=} S_\tau^{-1} X^T (X\beta^* + \varepsilon)$$

$$= S_\tau^{-1} (X^T X \beta^* + X^T \varepsilon)$$

$$\stackrel{S}{=} S_\tau^{-1} (S\beta^* + X^T \varepsilon)$$

$$= S_\tau^{-1} S\beta^* + S_\tau^{-1} X^T \varepsilon$$

$$\hat{\beta}_\tau \text{ is random only through } \varepsilon : \mathbb{E}[\varepsilon] = 0 \Rightarrow \mathbb{E}[\overbrace{S_\tau^{-1} X^T \varepsilon}^{\text{constant}}] = S_\tau^{-1} X^T \mathbb{E}[\varepsilon] = 0$$

$$\Rightarrow \mathbb{E}[\hat{\beta}_\tau] = \mathbb{E}[\underbrace{S_\tau^{-1} S\beta^*}_{\text{constant}}] + \mathbb{E}[S_\tau^{-1} X^T \varepsilon] = S_\tau^{-1} S\beta^*$$

$$\text{Cov}[\hat{\beta}_\tau] = \text{Cov}[\underbrace{S_\tau^{-1} X^T \varepsilon}_{\text{constant}}]$$

$$= S_\tau^{-1} X^T \text{Cov}[\varepsilon] (S_\tau^{-1} X^T)^T$$

$$= S_\tau^{-1} X^T \text{Cov}[\varepsilon] X S_\tau^{-1}$$

$$= S_\tau^{-1} X^T (\sigma^2 I_D) X S_\tau^{-1}$$

$$= S_\tau^{-1} (\sigma^2 X^T I_D X) S_\tau^{-1}$$

$$= S_\tau^{-1} S S_\tau^{-1} \sigma^2$$